
MICROECONOMY



*Prices, Markets, Costs and Rewards,
explained from first principles*

A TEXTBOOK FOR FIRST-YEAR STUDENTS

BIJOY?!



MICROECONOMY
BIJOY?!

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ANSWERS TO THE REVIEW QUESTIONS

CHAPTER I

1. Because every person and every country has limited resources compared with wants, whether rich or poor.
2. Micro: why does a loaf cost Tk 30? Macro: why are prices across the nation rising?
3. Normative. It says what ought to be and rests on values; facts alone cannot settle it.

CHAPTER II

1. The extra satisfaction a person gets from one more unit of a good.
2. 60 minus 50 = 10 utils.
3. Units of the same kind and reasonable size; no change in taste, income or prices.

CHAPTER III

1. When the price of a good rises, quantity demanded falls, other things equal.
2. Any two of: income, tastes, prices of related goods, expectations, number of buyers.
3. A basic good for which demand rises when its price rises.

CHAPTER IV

1. The percentage change in quantity demanded divided by the percentage change in price.
2. %change Q = -25%; %change P = 10%; Ed = 2.5, elastic.
3. The two goods are complements.

CHAPTER V

1. The difference between what a buyer would pay and what is paid.
2. They slope downward and are convex to the origin; they never cross.
3. 10 buns.

CHAPTER VI

1. When price rises, quantity supplied rises, other things equal.
2. Cheaper inputs, better technology, a subsidy, more sellers.
3. Exactly 1 at every point.

CHAPTER VII

1. The price at which quantity demanded equals quantity supplied.
2. A surplus; price falls.
3. $100 - 2P = 3P - 20$, so $120 = 5P$ and $P = 24$.

CHAPTER VIII

1. The relationship between inputs and the greatest output they can produce.
2. Diminishing returns: one input varies, others fixed (short run). Returns to scale: all inputs vary together (long run).
3. Decreasing returns to scale.

CHAPTER IX

1. The value of the best alternative given up.
2. The owner's time and the owner's own capital.
3. Rise, because MC is above AC.

CHAPTER X

1. The features of a market, such as number of sellers, product type and entry, that decide how firms behave.
2. Produce where price equals marginal cost, with MC rising.
3. The making of goods that are similar but seen by buyers as different.

CHAPTER XI

1. A payment to a factor in fixed supply, above what is needed to keep it in use.
2. Nominal wage is in money; real wage is what that money can buy.
3. Accounting profit 3,000; economic profit 1,800.

GLOSSARY AND INDEX

Each entry gives the term, its meaning, and the chapter and page of first use.

Accounting profit: total revenue minus explicit costs only. *Ch. 11, p. 46.*

Budget line: all combinations of two goods that the consumer's income can buy at given prices. *Ch. 5, p. 22.*

Complementary goods: goods used together, such as bread and butter. *Ch. 3, p. 14.*

Consumer's surplus: the difference between what a buyer is willing to pay and what the buyer actually pays. *Ch. 5, p. 21.*

Consumption: the use of goods and services to satisfy wants. *Ch. 5, p. 22.*

Cross elasticity of demand: the percentage change in quantity of one good divided by the percentage change in the price of another good. *Ch. 4, p. 18.*

Demand: the quantity of a good that buyers are willing and able to buy at each price in a given period. *Ch. 3, p. 13.*

Differential surplus: the extra produce of better land over the worst land in use. *Ch. 11, p. 45.*

Economic law: a statement of the usual relation between economic events, true when other things stay equal. *Ch. 1, p. vi.*

Economic profit: total revenue minus all costs, explicit and implicit. *Ch. 11, p. 46.*

Economic rent: the payment to a factor in fixed supply, above what is needed to keep it in use. *Ch. 11, p. 45.*

Economic system: the way a society organises the choices of what, how and for whom to produce. *Ch. 1, p. v.*

Economics: the study of how people choose to use limited resources to satisfy wants. *Ch. 1, p. iv.*

Elastic demand: quantity changes by a larger percentage than price. *Ch. 4, p. 17.*

Equilibrium: a state in which quantity demanded equals quantity supplied, so there is no pressure for change. *Ch. 7, p. 29.*

Exceptional supply curve: a supply curve that does not slope upward in the usual way. *Ch. 6, p. 26.*

Explicit cost: a cost that needs a payment to someone outside the firm. *Ch. 9, p. 37.*

Fixed cost: a cost that does not change with output in the short run. *Ch. 9, p. 37.*

Giffen good: a very basic good for which demand rises when its price rises. *Ch. 3, p. 14.*

Implicit cost: the value of resources the owner supplies, for which no payment is made. *Ch. 9, p. 37.*

Income elasticity of demand: the percentage change in quantity demanded divided by the percentage change in income. *Ch. 4, p. 18.*

Indifference curve: a line showing all combinations of two goods that give a consumer equal satisfaction. *Ch. 5, p. 22.*

Inelastic demand: quantity changes by a smaller percentage than price. *Ch. 4, p. 17.*

Interest: the payment for the use of borrowed money over a period of time. *Ch. 11, p. 46.*

Law of demand: when the price of a good rises, the quantity demanded falls, other things equal. *Ch. 3, p. 13.*

Law of diminishing marginal utility: as a person consumes more units, the extra satisfaction from each further unit falls. *Ch. 2, p. ix.*

Law of diminishing returns: when one input is added to fixed inputs, the extra output from each further unit eventually falls. *Ch. 8, p. 34.*

Law of supply: when the price of a good rises, the quantity supplied rises, other things equal. *Ch. 6, p. 25.*

Least-cost combination: the mix of inputs that produces a given output at the lowest total cost. *Ch. 9, p. 38.*

Long-run average cost curve: the lowest cost of each output when all inputs can change. *Ch. 9, p. 38.*

Macroeconomics: the study of the whole economy, such as total output, prices and jobs. *Ch. 1, p. v.*

Marginal utility: the extra satisfaction from one more unit. *Ch. 2, p. viii.*

Market: any arrangement that brings buyers and sellers of a good together to trade. *Ch. 7, p. 29.*



CHAPTER THE FIRST

WHAT ECONOMICS IS ABOUT

Why can nobody have everything?

The chapter begins with one oven and two choices. It shows why choice is unavoidable, defines economics and its two branches, and explains what an economic law is.

On 1 January 2024 the owner of a small bakery in Dhaka opens for the first time. She has one oven and eight hours a day to use it. In one day the oven can bake 400 loaves of bread or 80 cakes. It cannot bake both at full strength. Which should she bake? Any answer gives something up. Baking cakes means fewer loaves, and baking loaves means no cakes. This small problem is the whole of economics in miniature.

Start with a question you can almost answer.

CASE FILE: RUPALI BAKERY (ILLUSTRATIVE)

1 Jan 2024: opens in Mirpur, Dhaka. One oven, eight oven hours a day. Output choice per day: 400 loaves or 80 cakes, or a mixture in the same ratio (1 cake = 5 loaves). The owner is the only decision-maker.

THE MEANING OF ECONOMICS

Economics means the study of how people choose to use limited things to satisfy their wants. A **want** is anything a person would like to have, such as food, a phone or a holiday. A **resource** is anything used to make goods and services, such as land, labour, machines and money. The baker's oven hours are a resource. Her customers' wish for bread and cake is a want.

The central fact is **scarcity**. Scarcity means that resources are limited while wants are without end. Scarcity does not mean poverty. A rich country and a poor country both face it. Even a person with great wealth has only twenty-four hours in a day. Because of scarcity, every choice has a price, and that price is what we give up.

Writers have defined economics in several ways. Adam Smith (1776) described it as an inquiry into the nature and causes of the wealth of nations. Alfred Marshall (1890) called it a study of mankind in the ordinary business of life, and linked it to human welfare. Lionel Robbins (1932) wrote that it studies human behaviour as a relation between ends and scarce means that have alternative uses. Most modern teachers prefer the last, because it applies to every choice, not only to money.

The three definitions differ in their focus. Smith's focuses on wealth, Marshall's on welfare, and Robbins's on scarcity and choice. Critics said Smith

Economics the study of how people choose to use limited resources to satisfy wants.

The idea: choice under limits.

Scarcity resources are limited while wants are not.

Scarcity is the root of all economic problems.

Three famous definitions.

Each definition looked at a different part.

Nominal wage the wage paid in money.

Real wage what the money wage can buy.

Real wages count what buys.

Why wages vary.

How wages are determined.

Interest the payment for the use of borrowed money over a period of time.

Interest is the price of waiting.

Economic profit total revenue minus all costs, explicit and implicit.

Accounting profit total revenue minus explicit costs only.

Two ways to count profit.

WAGES

The **nominal wage** is the wage in money. The **real wage** is what that money can buy, found by correcting for prices. The formula is: real wage = nominal wage divided by the price index, times 100.

WORKED EXAMPLE. Nominal and real wage

GIVEN	A baker's wage rises from Tk 800 to Tk 880. The price index rises from 100 to 120.
FIND	The new real wage, and whether the worker gained.
FORMUL A	Real wage = nominal wage / price index x 100.
STEPS	Real wage = 880 / 120 x 100 = 733.3 (in base-year taka). Before: 800 / 100 x 100 = 800.
ANSWER	The real wage fell from Tk 800 to Tk 733.
CHECK	Check: wages rose 10% but prices rose 20%, so buying power must have fallen. It did.

Wages differ for reasons that are simple to state. Skilled work needs training, which costs time and money. Dangerous or unpleasant work needs extra pay. Some jobs are steady and others are not. Trade unions and law may raise some wages. Wages differ between regions because of different costs of living and different supply of labour. A job that gives status may pay less, because people want it for other reasons.

In a competitive labour market, wages are set by demand and supply. The employer's demand rests on **marginal productivity**: a worker is hired if the value of the extra output is at least the wage. In the bakery, each loaf adds Tk 15 of value after materials. The fourth baker adds 60 x 15 = Tk 900, above the wage of 800, so is hired. The fifth adds 40 x 15 = Tk 600, below 800, so is not. Labour supply and bargaining also matter, so the result may differ.

INTEREST

Interest means the payment for the use of borrowed money for a period. Lenders give up spending now and take risk, and interest rewards them. In the **loanable funds** view, the interest rate is set where the supply of savings meets the demand for loans. Keynes (1936) added that people hold money for safety and convenience, so interest is the price of giving up that liquidity. A loan of Tk 100,000 at 10 per cent costs Tk 10,000 a year. If prices rise 6 per cent, the **real interest rate** is about 10 - 6 = 4 per cent.

PROFIT

Accounting profit is revenue minus explicit costs. **Economic profit** is revenue minus all costs, including implicit ones. Using the bakery's figures at 500 loaves: revenue Tk 15,000 less explicit costs Tk 12,000 gives an accounting profit of Tk 3,000. Subtract the implicit costs of Tk 1,500 and the economic profit is Tk 1,500. **Normal profit** is the reward needed to keep the owner in business; it is already counted among the implicit costs.

from *makros*, “large”. It asks why prices rise across the country, or why total output grows slowly.

MICRO AND MACRO COMPARED

Point of difference	Microeconomics	Macroeconomics
Unit studied	One household, firm or market	The whole economy
Main question	How is price set?	What decides total output?
Main tools	Demand, supply, cost	Income, spending, money
Example	The price of bread	The rate of inflation

The two branches are not rivals. Total output is the sum of many single choices, and each firm's choices depend on the state of the whole economy. A fall in national income reduces bread sales. Still, you must learn the parts before the whole, and that is why this book starts with micro.

THE SCOPE AND USE OF MICROECONOMICS

The **scope** of a subject means the range of topics it covers. Microeconomics covers the theory of demand, which explains buyers; the theory of supply and production, which explains sellers; the theory of the firm and of market structures, which explains competition and monopoly; and the theory of distribution, which explains rent, wages, interest and profit. Welfare economics, which asks whether the result is good for society, is a further part.

Its objectives are practical. It explains how prices are set, so that firms can decide what to produce and at what price. It shows how resources are shared among uses. It helps governments judge taxes, subsidies and price controls. It also helps a consumer plan spending. A firm that knows its costs and its customers decides better.

In Bangladesh the subject has particular value. The country has many small farms, workshops and shops, so market prices matter to millions of people. Planners need to judge the effect of taxes and price rules on a developing economy. Support for firms comes from institutions such as Bangladesh Bank, the Bangladesh Small and Cottage Industries Corporation (BSCIC) and the Bangladesh Competition Commission. These bodies rely on microeconomic ideas when they set rules for credit, small enterprise and fair competition.

ECONOMIC LAWS, SCIENCE AND ART

An **economic law** means a statement of how people usually behave in economic matters. An example: when the price of a good rises, buyers tend to buy less. Such laws differ from the laws of physics in four ways. They state tendencies, not certainties, because human beings choose. They hold only if other things stay the same, a condition known by the Latin phrase *ceteris paribus*. They are less exact, as they cannot be tested in a laboratory. And they apply to groups of people more than to one person.

Economics contains two kinds of statement. A **positive** statement says what is, and facts can test it: “A higher bread price reduces bread sales.” A **normative** statement says what ought to be, and it rests on values: “The government should fix the price of bread.” Positive economics explains. Normative economics judges. Good reasoning keeps the two apart, because facts can settle the first but not the second.

The two are linked.

Scope means what the subject covers.

Why the study is useful.

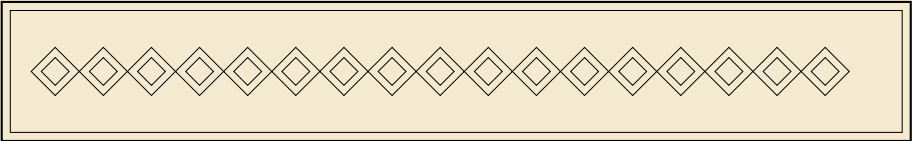
Prospects in Bangladesh.

Economic law a statement of the usual relation between economic events, true when other things stay equal.

A law states a tendency, not a certainty.

Positive statement a statement about what is, which facts can test.

Positive: what is. Normative: what ought to be.



CHAPTER THE SECOND

UTILITY AND SATISFACTION

Why does the fifth bun taste less sweet?

The chapter begins with a puzzle about water and diamonds. It defines utility, builds the law of diminishing marginal utility from a hungry student’s buns, and shows where the law holds and where it fails.

Water is cheap, though without it we die. Diamonds are costly, though we can live without them. Why? Why does the more useful good cost less? The puzzle puzzled thinkers for a century. The answer needs a new idea: the value we place on a good depends on the extra satisfaction we gain from one more unit of it. We build that idea step by step.

Start with a question you can almost answer.

UTILITY, TOTAL AND MARGINAL

Utility means the satisfaction a person gets from using a good or service. The word does not mean usefulness in a moral sense. A cigarette has utility for a smoker, though it harms health. Utility depends on the person, the time and the place. A cold drink gives more utility on a hot day than on a cold one.

Two measures matter. **Total utility** (TU) means the whole satisfaction from all the units consumed. **Marginal utility** (MU) means the extra satisfaction from one more unit. In words, marginal utility equals the change in total utility divided by the change in the number of units. The word *marginal* means “at the edge”, that is, the last or the next unit.

Utility the satisfaction a person gets from using a good.
Utility is satisfaction.

Total utility the whole satisfaction from all the units consumed.
Marginal utility the extra satisfaction from one more unit.
Two measures: total and marginal.

WORKED EXAMPLE. Total and marginal utility

GIVEN	A hungry student eats buns. TU after 1, 2, 3, 4, 5, 6 buns is 20, 36, 48, 56, 60, 60 utils.
FIND	The marginal utility of each bun.
FORMUL	MU of the nth bun = TU(n) minus TU(n-1).
A	
STEPS	MU1 = 20 - 0 = 20. MU2 = 36 - 20 = 16. MU3 = 48 - 36 = 12. MU4 = 56 - 48 = 8. MU5 = 60 - 56 = 4. MU6 = 60 - 60 = 0.
ANSWER	The marginal utilities are 20, 16, 12, 8, 4, 0 utils.
CHECK	Add them: 20+16+12+8+4+0 = 60, the total utility. The check holds.

A **util** is an imaginary unit used to count satisfaction. No machine measures it. Is satisfaction measurable at all? Early economists, such as Jevons and Marshall, said yes: a person can state how much they would pay, and that measures it in money. This is the **cardinal** view. Later writers, such as Hicks,

A unit of measure.

Equilibrium of the firm.

A firm that wishes to earn the most profit expands output as long as the extra unit adds more to revenue than to cost. It stops where MR equals MC, with MC rising. For the competitive firm, MR equals price, so the rule is $P = MC$. In the short run the firm may earn a profit, break even or make a loss. It continues to produce in the short run if price covers at least average variable cost.

WORKED EXAMPLE. Short-run profit of the firm

GIVEN	Market price Tk 30. Costs from Chapter 9: at 500 loaves, $TC = Tk\ 12,000$ and $MC = Tk\ 29$.
FIND	The output that maximises profit, and the profit.
FORMUL A	Rule: $P = MC$. Profit = $(P - AC) \times Q$.
STEPS	At 400 loaves MC is Tk 19, below the price; the firm gains by baking more. At 500 loaves MC is Tk 29, just below Tk 30, so 500 is the best output on the table. $AC = 12,000 / 500 = 24$. Profit = $(30 - 24) \times 500 = 6 \times 500 = 3,000$.
ANSWER	The firm bakes 500 loaves and earns Tk 3,000 a day.
CHECK	Check: revenue $30 \times 500 = 15,000$; cost 12,000; the difference is 3,000. It matches.

The industry in the short and long run.

The **industry** means all firms that make the same good. In the short run, the industry is in equilibrium when market demand equals the sum of the supplies of all firms, which fixes the price each firm accepts. In the long run, profit attracts new firms. Their entry raises supply and lowers price. Entry continues until price equals the lowest AC, and each firm earns only **normal profit**, the minimum reward needed to keep it in business. A loss has the opposite effect: firms leave, supply falls, and price rises. The industry reaches long-run equilibrium when no firm wishes to enter or leave.

MONOPOLY

Monopoly a market with a single seller of a good that has no close substitute.

One seller sets the price.

Monopoly means a market with only one seller and no close substitute. A **pure monopoly** is the extreme case. A **bilateral monopoly** has one seller and also only one buyer, such as a single union facing a single employer. The monopolist faces the whole market demand, so to sell more it must lower the price on all units. Its marginal revenue is therefore below price. It maximises profit where MR equals MC, then charges the highest price buyers will pay for that output, which is above MC.

MONOPOLISTIC COMPETITION

Monopolistic competition a market with many sellers of similar but not identical goods, and free entry.

Many sellers, each with a small monopoly.

Monopolistic competition means a market with many sellers whose goods are similar but not identical, with free entry. Chamberlin and Robinson described it in 1933. **Product differentiation** is its typical feature. It means that each firm makes a version of the good that buyers see as different, through brand, quality, taste or location. Rupali’s loaf, with its own recipe and shop, is one example. Differentiation gives each firm a small monopoly and a downward-sloping demand curve.

In the short run, the firm acts like a monopolist: it sets MR equal to MC and may earn a profit. In the long run, profit invites new firms with close substitutes. The firm’s demand shifts left and becomes flatter, until the demand curve just touches the AC curve. Price equals AC, and the firm earns only normal profit. Yet the output is below the point of lowest cost. This is called **excess capacity**.

Short and long run.

SUMMARY. Utility is satisfaction. Total utility is the whole; marginal utility is the extra from one more unit. As consumption rises, marginal utility falls. The law needs stated conditions, and the apparent exceptions break those conditions. Value follows marginal utility.

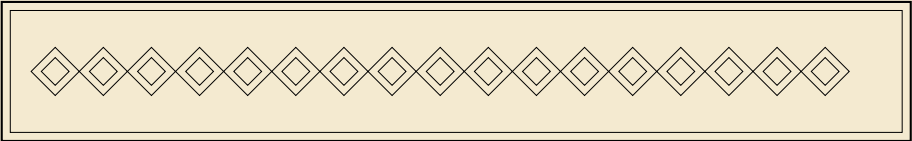
THINK IT THROUGH. Using the survey, how many buns would the customer buy at Tk 5? Explain by comparing price with marginal utility.

WHERE PEOPLE GO WRONG. “Diminishing utility means total utility falls.” It does not. Total utility rises more slowly, and it falls only when marginal utility becomes negative.

IN HISTORY. In 1738 Daniel Bernoulli argued that an extra ducat means less to a rich man than to a poor man. He used it to explain risky choices. The marginal revolution of 1871-74 turned this insight into the basis of value theory.

REVIEW QUESTIONS

1. Define marginal utility in one sentence.
2. If TU is 30, 50, 60, 62, what is MU of the third unit?
3. Name two conditions the law needs.



CHAPTER THE THIRD

THE THEORY OF DEMAND

Why do we buy less when prices rise?

The chapter begins with a bakery price change. It defines demand, states the law, builds the demand curve from a schedule, and examines the exceptions and the forces that shift the curve.

On 1 March 2024 the bakery raises the price of a loaf from Tk 25 to Tk 30. By the end of the week, fewer loaves are sold. Was this a surprise? Why does a higher price so often meet lower sales? The answer is the oldest regularity in economics, and it is the first half of the story of every market.

Start with a question you can almost answer.

DEMAND AND ITS LAW

Demand means the quantity of a good that buyers are willing and able to buy at each possible price in a given period. Both parts matter. A wish without money is not demand. A **demand schedule** is a table of prices and quantities. A **demand curve** is the same information drawn as a line.

Demand the quantity of a good that buyers are willing and able to buy at each price in a given period.

Demand needs both wish and ability to pay.

DAILY DEMAND FOR LOAVES IN THE NEIGHBOURHOOD

Price per loaf (Tk)	20	25	30	35	40
Quantity demanded	180	150	120	90	60

The table shows the **law of demand**. It states that, when the price of a good rises, the quantity demanded falls, and when the price falls, the quantity rises, if other things stay the same. Three reasons lie behind it. As price rises, we buy cheaper substitutes (the substitution effect). A higher price also reduces our real buying power (the income effect). And by the law of diminishing marginal utility, we will pay less for each extra unit.

Law of demand when the price of a good rises, the quantity demanded falls, other things equal.

Price up, quantity down.

WORKED EXAMPLE. Cost per loaf

GIVEN	At 300 loaves, FC = Tk 1,500 and VC = Tk 5,700. Going from 300 to 400 loaves raises VC to Tk 7,600.
FIND	AC at 300 loaves, and MC of the step from 300 to 400.
FORMUL A	$AC = TC / Q$. $MC = \text{change in } TC / \text{change in } Q$.
STEPS	$TC = 1,500 + 5,700 = 7,200$. $AC = 7,200 / 300 = 24$. At 400: $TC = 1,500 + 7,600 = 9,100$. Change in $TC = 9,100 - 7,200 = 1,900$. $MC = 1,900 / 100 = 19$.
ANSWER	$AC = \text{Tk } 24$ a loaf; $MC = \text{Tk } 19$ a loaf.
CHECK	Check: MC (19) is below AC (24), so AC should be falling as output rises to 400. In the table it falls to 22.8.

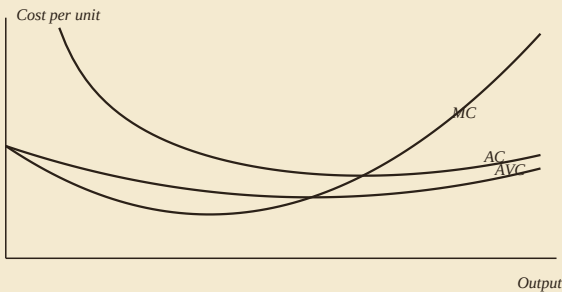


FIG. 10. Short-run cost curves. MC cuts both AVC and AC at their lowest points.

The pattern.

Look at the curves. AC and AVC fall at first and then rise, so each is U-shaped. AC falls at first because fixed cost is spread over more loaves. It rises later because of diminishing returns (Chapter 8). MC rises faster than the others and cuts them at their lowest points. A good rule: if MC is below the average, the average falls; if MC is above, the average rises. A marginal figure pulls the average in its direction.

SHORT RUN AND LONG RUN

Long-run average cost curve the lowest cost of each output when all inputs can change.

In the long run, no cost is fixed.

In the short run, the bakery has one oven, so it chooses a cost curve for that size. In the long run, it can choose a larger or smaller oven. The **long-run average cost curve** (LAC) joins the lowest points available from all the short-run curves, so it touches each of them from below. It is U-shaped as well: it falls with economies of scale and rises with diseconomies of scale (Chapter 8).

LEAST-COST COMBINATION

Least-cost combination the mix of inputs that produces a given output at the lowest total cost.

Spend each taka where it buys most.

A firm can often make the same output with different mixes of labour and machines. The **least-cost combination** is the mix that gives a target output at the lowest cost. The rule is: the extra output per taka should be equal for every input. In words, marginal product of labour divided by the wage must equal marginal product of capital divided by its hire price.

demand rises when many people are seen buying a popular phone. Expectations also matter. If buyers expect the price to rise further, they may buy more today. In each case the demand curve may slope upward over part of its range.

CASE FILE: DEMAND SCHEDULE, 1 MARCH 2024

Daily quantity demanded in the neighbourhood: Tk 20 gives 180 loaves, Tk 25 gives 150, Tk 30 gives 120, Tk 35 gives 90, Tk 40 gives 60. In equation form, $Q = 300 - 6P$. Check: $P = 30$ gives $Q = 300 - 180 = 120$.

SUMMARY. Demand is the quantity buyers are willing and able to buy at each price. The law says price and quantity move in opposite directions. A price change moves us along the curve; a change in income, taste, related prices or number of buyers shifts it. Giffen goods, snob and bandwagon effects are exceptions.

THINK IT THROUGH. From $Q = 300 - 6P$, find the quantity demanded at Tk 22. Then say whether a rise in the price of flour shifts this demand curve.

WHERE PEOPLE GO WRONG. “A fall in price shifts the curve to the right.” It does not. A price change moves us along the same curve; only other causes shift it.

IN HISTORY. Alfred Marshall’s *Principles* (1890) drew demand with price on the vertical axis and quantity on the horizontal. Economists have kept that convention for more than a century.

REVIEW QUESTIONS

1. State the law of demand.
2. Name two causes of a shift in demand.
3. What is a Giffen good?



CHAPTER THE FOURTH

ELASTICITY OF DEMAND

How much does demand react to a price change?

The chapter begins with two price rises that bring different results. It defines elasticity, measures price, income and cross elasticity, names the five forms, and lists the factors behind each.

Two shops each raise their price by 10 per cent. The bread shop loses almost no customers. The cake shop loses a third of them. Why do the same price rises give such different results? The law of demand tells us the direction of the change. It does not tell us the size. To measure size, we need a new tool.

Start with a question you can almost answer.

PRICE ELASTICITY OF DEMAND

Price elasticity of demand means the percentage change in quantity demanded divided by the percentage change in price. In symbols, $E_d = (\% \text{change in } Q) / (\% \text{change in } P)$. Percentage change means the change divided by the starting value, times 100. Because price and quantity move in opposite directions, the answer is negative. We usually drop the minus sign and speak of the size.

Price elasticity of demand the percentage change in quantity demanded divided by the percentage change in price. *Elasticity measures how strongly buyers react.*

WORKED EXAMPLE. Price elasticity from a schedule

GIVEN	From the demand schedule of Chapter 3: at Tk 30 quantity is 120; at Tk 35 quantity is 90.
FIND	The price elasticity of demand for the rise from Tk 30 to Tk 35.
FORMULA	$E_d = (\% \text{change in } Q) / (\% \text{change in } P)$.
STEPS	Change in $Q = 90 - 120 = -30$. $\% \text{change in } Q = -30/120 \times 100 = -25\%$. Change in $P = 35 - 30 = 5$. $\% \text{change in } P = 5/30 \times 100 = 16.7\%$. $E_d = -25 / 16.7 = -1.5$.
ANSWER	The size of elasticity is 1.5. Demand is elastic.
CHECK	Check: the percentage fall in quantity (25) is larger than the percentage rise in price (16.7), so the ratio must exceed 1. It does.

Now take a fall in price from Tk 25 to Tk 20. Quantity rises from 150 to 180. The change in Q is +30, which is 20 per cent of 150. The change in P is -5, which is 20 per cent of 25. So $E_d = 20 / (-20) = -1$. Take care: a common slip is to use the new value instead of the starting value, which gives a different answer. Always divide by the starting value, or use the midpoint if your teacher asks.

There are five forms. If E_d is greater than 1, demand is **elastic**: quantity changes by a larger percentage than price. If E_d is less than 1, demand is **inelastic**: quantity changes by a smaller percentage. If E_d equals 1, demand is **unitary elastic**. At the two extremes, demand is **perfectly inelastic** ($E_d = 0$,

A second example, with messier numbers.

Elastic demand quantity changes by a larger percentage than price.
Inelastic demand quantity changes by a smaller percentage than price.
Five forms, from zero to infinity.

Law of diminishing returns when one input is added to fixed inputs, the extra output from each further unit eventually falls.
Eventually each extra worker adds less.

The table shows the **law of diminishing returns**. It states that, when more of one input is added to fixed amounts of the others, the marginal product of the added input eventually falls. Here it begins after the third baker, whose MP is 80. Turgot noted the effect in 1767, and Malthus, West and Ricardo stated it in 1815. Ricardo applied it to farmland.

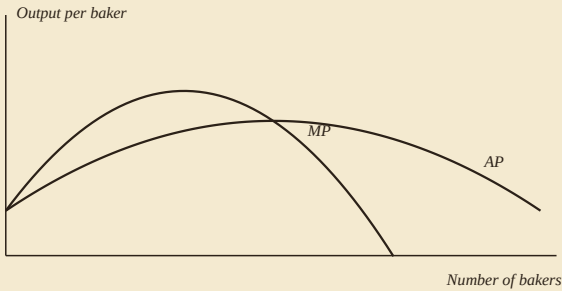


FIG. 9. Marginal product (MP) rises, then falls and cuts the average product (AP) curve at its highest point.

Why it happens.

The cause is simple. The oven is fixed. At first, extra bakers share tasks and use the oven better, so MP rises. Past a point, bakers crowd the kitchen and wait for the oven, so each adds less. The law holds if the extra units are of equal skill, if technology is unchanged and if at least one input is fixed. Output is in three stages: rising MP, falling but positive MP (where a sensible firm operates), and negative MP.

RETURNS TO SCALE

Returns to scale the change in output when all inputs are changed in the same proportion.
The long-run cousin of diminishing returns.

Returns to scale means the change in output when every input is changed in the same proportion. It is a long-run idea, since all inputs vary. Diminishing returns is a short-run idea, since one input is fixed. Do not confuse them. If output rises by a larger percentage than inputs, there are **increasing returns to scale**. If it rises by the same percentage, they are **constant**. If it rises by a smaller percentage, they are **decreasing**.

WORKED EXAMPLE. Types of returns to scale

GIVEN	Units of all inputs: 1, 2, 4, 8. Output: 100, 250, 500, 700.
FIND	Whether returns are increasing, constant or decreasing at each step.
FORMUL	%change in output compared with %change in inputs.
A	
STEPS	Step 1 to 2: inputs +100%; output 100 to 250 is +150%. Step 2 to 4: inputs +100%; output 250 to 500 is +100%. Step 4 to 8: inputs +100%; output 500 to 700 is +40%.
ANSWER	Increasing, then constant, then decreasing returns.
CHECK	Check: +150% exceeds +100%, +100% equals +100%, and +40% is below +100%. Each label fits.

Why returns to scale change.

At first, a larger firm gains economies: workers specialise, big machines work well, bulk buying lowers costs. Later, size brings strain: managers lose touch, orders move slowly and costs rise. Smith’s pin factory (1776) shows the first effect. Ford’s Model T line shows it too: the price fell from \$850 in 1908 to under \$300 by the mid-1920s, as assembly-line output grew.

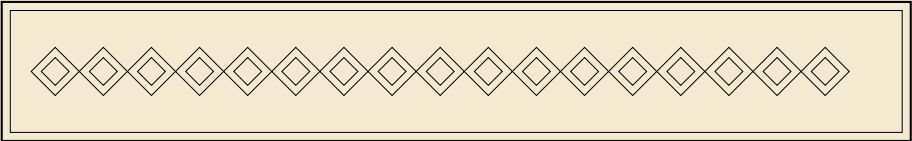
THINK IT THROUGH. A firm sells 200 units at Tk 10. It raises the price to Tk 11 and sells 150. Find the elasticity and say what happens to revenue.

WHERE PEOPLE GO WRONG. “A steep curve and an inelastic curve are the same.” Slope and elasticity are different, because elasticity uses percentages and slope uses units. A straight line has one slope but a different elasticity at each point.

IN HISTORY. Marshall introduced the measure of price elasticity of demand in 1890. He wanted a number that did not depend on units, so that the demand for wheat and the demand for cloth could be compared.

REVIEW QUESTIONS

- 1. Define price elasticity of demand.
- 2. Q = 200, P = 10; Q falls by 50 when P rises by 1. Find Ed.
- 3. What does a cross elasticity of -0.8 show?



CHAPTER THE FIFTH

CONSUMER SURPLUS AND INDIFFERENCE CURVE

Why does a bargain feel like a gift?

The chapter begins with a customer who pays less than she was willing to pay. It measures this gain as consumer surplus, then shows how indifference curves describe choice without counting utils.

In Chapter 2 a customer was ready to pay Tk 20 for her first bun. The price is only Tk 10. She walks away pleased. Where did the pleasure come from, and can we count it in taka? The answer gives economists a way to judge whether a market makes people better off.

Start with a question you can almost answer.

CONSUMER'S SURPLUS

Consumer's surplus means the difference between the most a buyer is willing to pay and the price actually paid. Dupuit (1844) first described the idea, and Marshall (1890) developed it. It rests on diminishing marginal utility: because each extra unit is worth less, only the last unit is worth exactly the price.

Consumer's surplus the difference between what a buyer is willing to pay and what the buyer actually pays.
The gain from paying less than you would.

WORKED EXAMPLE. Consumer's surplus from the survey

GIVEN	The customer values the buns at Tk 20, 16, 12, 8 and 4. The price is Tk 10.
FIND	Her consumer's surplus.
FORMULA	Surplus on a unit = value minus price. Buy a unit only if value is at least the price.
STEPS	She buys the first three buns, valued at 20, 16 and 12. Surplus: $(20 - 10) + (16 - 10) + (12 - 10) = 10 + 6 + 2 = 18$.
ANSWER	Her consumer's surplus is Tk 18.
CHECK	Check: she paid $3 \times 10 = \text{Tk } 30$ for buns she valued at $20 + 16 + 12 = \text{Tk } 48$. The gap is $48 - 30 = 18$. It agrees.

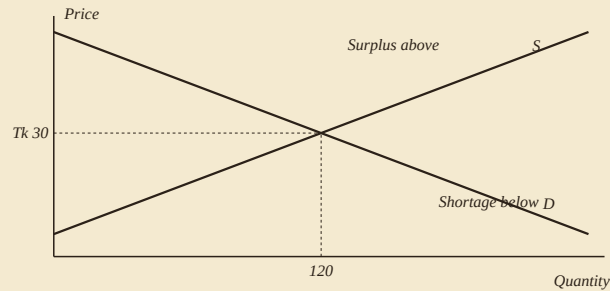


FIG. 8. The curves cross at the equilibrium. Above the price there is a surplus; below it, a shortage.

The forces at work.

Now see why other prices fail. At Tk 40, quantity supplied is 180 and quantity demanded is 60, so there is a **surplus** of 120 loaves. Sellers have unsold bread, and they cut the price. At Tk 20, quantity demanded is 180 and quantity supplied is 60, so there is a **shortage** of 120. Buyers compete, and the price rises. In both cases the price moves toward Tk 30. No one sets it. It emerges from many separate choices.

When a curve shifts.

Suppose a change in tastes adds 60 loaves to demand at every price, so $Q_d = 360 - 6P$. Then $360 - 6P = 6P - 60$, so $420 = 12P$ and $P = 35$. Quantity is $Q = 6 \times 35 - 60 = 150$. A rise in demand raises both price and quantity. A rise in supply does the opposite to price: it lowers price and raises quantity. Always find which curve moved, in which direction, and then read the new crossing.

PRICE IN DIFFERENT PERIODS

Time matters.

Marshall (1890) distinguished three periods. In the **market period**, supply is fixed, as with the morning's baked loaves, so demand alone sets the price. In the **short run**, firms can change output using their existing ovens, but cannot build new ones. In the **long run**, firms can enter and leave, and all inputs can change. Prices in the long run follow cost more closely, and prices in the short run follow demand more closely. Under other market types, as Chapter 10 shows, a firm may influence the price itself.

Is a lower price always good?

Not always. A lower price helps buyers, and consumer's surplus rises. But if the price falls below the cost of production, firms leave, supply shrinks and shortages follow. A government price ceiling set below equilibrium creates exactly the shortage described above. A very low price may also signal poor quality, or may be used by a large firm to drive out smaller rivals. A lower price is good for buyers only when it reflects lower cost and when the sellers remain.

CASE FILE: MARKET TEST, 4-6 MARCH 2024

Monday Tk 40: demand 60, supply 180, surplus 120. Tuesday Tk 20: demand 180, supply 60, shortage 120. Wednesday Tk 30: demand 120, supply 120. Revenue at the equilibrium: $30 \times 120 = \text{Tk } 3,600$ a day.

SUMMARY. A market brings buyers and sellers together. The equilibrium price makes quantity demanded equal to quantity supplied. A surplus pulls price down and a shortage pushes it up. A shift in demand or supply creates a new equilibrium. Lower prices are not always good.

tea, or 6 cups and no buns, or any point between, such as 6 buns and 3 cups. In words: spending on buns plus spending on tea equals income.

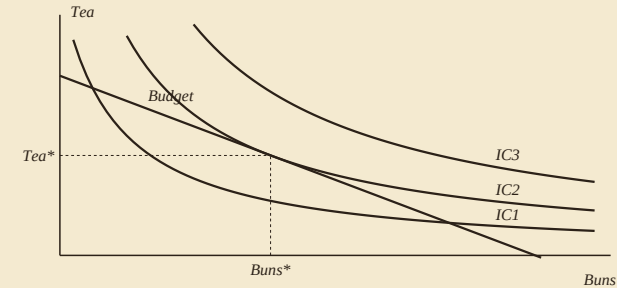


FIG. 6. The consumer chooses the point where the budget line just touches the highest curve she can reach.

The consumer reaches **equilibrium**, the point of best choice, where the budget line touches the highest indifference curve. At that point the two lines have the same slope. So the rate at which she is willing to swap goods (MRS) equals the rate at which the market lets her swap them (the price ratio). If the price of one good falls, the budget line turns outward, and she reaches a higher curve. The extra satisfaction can be valued in money, which links this method back to consumer's surplus.

Equilibrium of the consumer.

Hicks's method measures the same surplus without utils. After a price fall, the consumer reaches a higher indifference curve. Now ask how much money we must take from her income to push her back to her old curve. That sum is her gain from the price fall, and so her consumer's surplus in money. The method needs only a ranking of bundles, not a count of satisfaction.

Measuring surplus with indifference curves.

SUMMARY. Consumer's surplus is value minus price, summed over the units bought. Producer's surplus is its mirror. Indifference curves rank bundles without counting utils; the budget line shows what can be bought; the best choice is where the two touch.

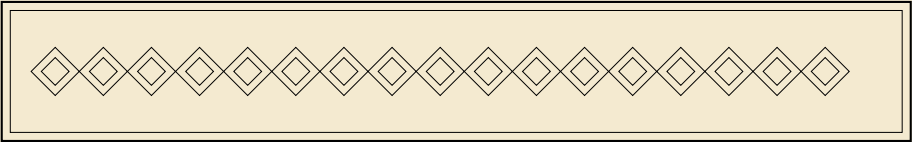
THINK IT THROUGH. With income Tk 120, a bun at Tk 10 and tea at Tk 20, can the consumer buy 4 buns and 4 cups? Show your working.

WHERE PEOPLE GO WRONG. "Indifference curves can cross if the goods are similar." They cannot. If they crossed, one bundle would be equal to two others that are not equal to each other.

IN HISTORY. In 1844 the French engineer Jules Dupuit asked how to value a public bridge when it charged a toll. He saw that users would have paid more than the toll, and he measured the difference. This became consumer's surplus.

REVIEW QUESTIONS

1. Define consumer's surplus.
2. Give two properties of indifference curves.
3. With income Tk 100, bun Tk 10, how many buns if nothing else is bought?



CHAPTER THE SIXTH

THE THEORY OF SUPPLY

Why do sellers offer more at a higher price?

The chapter begins with a baker who bakes more when bread sells for more. It states the law of supply, lists what shifts supply, notes the exceptions, and measures the response by elasticity of supply.

In the spring of 2024 the price of bread in the neighbourhood rises. Rupali Bakery bakes more, and two other bakeries do the same. A higher price gives sellers a reason to offer more. This is the second half of every market story, and it mirrors the story of buyers.

Start with a question you can almost answer.

SUPPLY AND ITS LAW

Supply means the quantity of a good that sellers are willing and able to offer for sale at each price in a given period. A **supply schedule** is a table of prices and quantities. A **supply curve** is the same table drawn as a line. A schedule gives separate numbers; a curve joins them into one picture.

Supply the quantity of a good that sellers are willing and able to offer at each price in a given period. Supply is the sellers' side.

DAILY SUPPLY OF LOAVES IN THE NEIGHBOURHOOD					
Price per loaf (Tk)	20	25	30	35	40
Quantity supplied	60	90	120	150	180

The table shows the **law of supply**. It states that, when the price of a good rises, the quantity supplied rises, and when the price falls, the quantity falls, if other things stay the same. Two reasons lie behind it. A higher price makes it more profitable to produce. And producers face rising cost per unit when they expand, so they need a higher price to cover it. In equation form, $Q_s = 6P - 60$.

Law of supply when the price of a good rises, the quantity supplied rises, other things equal. Price up, quantity up.